RAMBØLL

PROJECT
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PROJECT NO.
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Concrete Modulus at Age t (EN 1992-1-1 3.1.3)

EN 1992-1-1:2004 §3.1.2, §3.1.3

PREPARED TE
CHECKED —
APPROVED —

REV / DATE
0.1 · 2026-07-19

DRAFT

PURPOSE

Time-development of the concrete secant modulus of elasticity, from cement class and mean 28-day strength. Shared upstream of crack-width, deflection and bending-stiffness calculations.

1 Input Parameters

Symbol	Value	Unit	Description
s_{cem}	0.25	-	Cement class coefficient (0.20 R / 0.25 N / 0.38 S)
t	7	days	Concrete age at loading
f_{cm}	38	MPa	Mean compressive strength at 28 days (C30/37)
E_{cm}	32837	MPa	Secant modulus at 28 days (C30/37)

2 Calculation (all formulas from the shared library)

Concrete strength development coefficient at age t

$$\beta_{cc,t} = e^{s_{cem} \cdot \left(1 - \sqrt{\frac{28}{t}}\right)} = e^{0.25 \cdot \left(1 - \sqrt{\frac{28}{7}}\right)} = \mathbf{0.7788}$$

EN 1992-1-1:2004 §3.1.2(6), Eq. (3.2)

Mean concrete compressive strength at age t

$$f_{cm,t} = \beta_{cc,t} \cdot f_{cm} = 0.7788 \cdot 38 = \mathbf{29.59}\ \text{MPa}$$

EN 1992-1-1:2004 §3.1.2(6), Eq. (3.1)

Secant modulus of elasticity at age t

$$E_{cm,t} = E_{cm} \cdot \left(\frac{f_{cm,t}}{f_{cm}}\right)^{0.3} = 32837 \cdot \left(\frac{29.59}{38}\right)^{0.3} = \mathbf{30464}\ \text{MPa}$$

EN 1992-1-1:2004 §3.1.3(3), Eq. (3.5)

3 Verification

OK — **Modulus at early age does not exceed the 28-day value**

$E_{cm,t} = 30464\ \text{MPa} \le E_{cm} = 32837\ \text{MPa}$